

Supplementary Material document for “Focused median bias reduction”

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Supplementary material

This document provides the proof of Theorem 1, supplementary derivations and simulation results, and additional examples accompanying “Focused median bias reduction”. The repository <https://github.com/ikosmidis/fmedbr-supplementary> contains this document and scripts to reproduce all numerical results in the main text and supplementary material.

S1 Proof of Theorem 1

Write $W = W_1 - W_2$, where $W_1 \equiv W_1(\hat{\psi}; \theta) = K_2^{-1/2}(\hat{\psi} - \psi)$ and $W_2 \equiv W_2(\hat{\psi}; \theta) = K_2^{-1/2}B_{(\psi)}$, where $\psi = h(\theta)$. Then, W is asymptotically standard normal. Furthermore, $B_{(\psi)} = O(n^{-1})$, $K_2 = O(n^{-1})$, and $\rho_3 = O(n^{-1/2})$. Consider the estimator $\tilde{\psi}^{(a)}$ that results as the solution of the estimating equation $Z = 0$ with respect to ψ .

First, the roots of Z as a quadratic function of W are $O_p(n^{-1/2})$ and $O_p(n^{1/2})$. We focus only on the former root, which is the only one that remains compatible with (1) when $\rho_3 \rightarrow 0$. For that root, the term $\rho_3 W^2/6$ is $O_p(n^{-3/2})$, and solving $Z = 0$ with respect to ψ yields the solution

$$\tilde{\psi}^{(a)} = \hat{\psi} - B_{(\psi)} + \frac{1}{6} \frac{K_3}{K_2} + O_p(n^{-3/2}). \quad (\text{S1})$$

By the Cornish-Fisher inversion corresponding to the second-order Edgeworth expansion of W , the 1/2-quantile of W is $-\rho_3/6 + O(n^{-3/2})$, and, therefore, $P(Z \leq 0) = P(W \leq -\rho_3/6) + O(n^{-3/2})$. An Edgeworth expansion (see, for example, [Pace and Salvan, 1997](#), Section 10.4) gives

$$P\left(W \leq -\frac{\rho_3}{6}\right) = \Phi\left(-\frac{\rho_3}{6}\right) + \phi\left(-\frac{\rho_3}{6}\right) \left\{ \frac{\rho_3}{6} + O(n^{-3/2}) \right\},$$

where $\phi(\cdot)$ and $\Phi(\cdot)$ are the density and distribution function of the standard normal distribution, respectively. The error is of order $O(n^{-3/2})$ because the $O(n^{-1})$ term in the Edgeworth expansion is a linear combination with coefficients of order $O(n^{-1})$ of odd Hermite polynomials evaluated at

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$-\rho_3/6$. Using the expansions $\Phi(-\rho_3/6) = 1/2 - \phi(0)\rho_3/6 + O(n^{-3/2})$, and $\phi(-\rho_3/6) = \phi(0) + O(n^{-1})$, we obtain cancellation of the $O(n^{-1/2})$ terms, and, hence

$$P(Z \leq 0) = \frac{1}{2} + O(n^{-3/2}).$$

Assuming that Z is strictly decreasing in a neighbourhood of ψ , we get that the event $\{Z \leq 0\}$ has the same probability as the event $\{\tilde{\psi}^{(a)} \leq \psi\}$. As a result,

$$P(\tilde{\psi}^{(a)} \leq \psi) = \frac{1}{2} + O(n^{-3/2}).$$

S2 Derivation of κ_3 in Section 2.4

In this section we derive the expression of the asymptotic expansion (8) for the third cumulant of the maximum likelihood estimator $\hat{\psi} = h(\hat{\boldsymbol{\theta}})$ of the focus parameter, where $\hat{\boldsymbol{\theta}}$ is the maximum likelihood estimator of the parameter vector $\boldsymbol{\theta}$. In the following, in the first part we adopt the index notation and Einstein summation convention, with summation over repeated indices $r, s, t \in \{1, \dots, p\}$ (McCullagh, 2018; Pace and Salvan, 1997, Chapter 9). Let $\boldsymbol{\Delta} = \hat{\boldsymbol{\theta}} - \boldsymbol{\theta}$, and denote its r th component as $\Delta^r = \hat{\theta}^r - \theta^r$. Write

$$\dot{\psi}_r = \frac{\partial h(\boldsymbol{\theta})}{\partial \theta^r}, \quad \ddot{\psi}_{rs} = \frac{\partial^2 h(\boldsymbol{\theta})}{\partial \theta^r \partial \theta^s},$$

and denote by i^{rs} the (r, s) element of $\mathbf{i}^{-1}$. A second-order Taylor expansion of $\hat{\psi}$ around $\boldsymbol{\theta}$ gives

$$\hat{\psi} = \psi + \dot{\psi}_r \Delta^r + \frac{1}{2} \ddot{\psi}_{rs} \Delta^r \Delta^s + O_p(n^{-3/2}). \quad (\text{S2})$$

Taking expectations in (S2),

$$\mathbb{E}(\Delta^r) = O(n^{-1}), \quad \mathbb{E}(\Delta^r \Delta^s) = i^{rs} + O(n^{-2}),$$

imply

$$\mathbb{E}(\hat{\psi}) = \psi + \dot{\psi}_r \mathbb{E}(\Delta^r) + \frac{1}{2} \ddot{\psi}_{rs} i^{rs} + O(n^{-2}). \quad (\text{S3})$$

Subtracting (S3) from (S2) yields

$$\hat{\psi} - \mathbb{E}(\hat{\psi}) = A + B + O_p(n^{-3/2}),$$

where

$$A = \dot{\psi}_r \{\Delta^r - \mathbb{E}(\Delta^r)\}, \quad B = \frac{1}{2} \ddot{\psi}_{rs} (\Delta^r \Delta^s - i^{rs}).$$

Since $A = O_p(n^{-1/2})$ and $B = O_p(n^{-1})$, it follows that

$$K_3 \equiv \text{cum}_3(\hat{\psi}) = C_3 + O(n^{-3}),$$

where

$$C_3 = \mathbb{E}(A^3) + 3\mathbb{E}(A^2 B). \quad (\text{S4})$$

The first term in (S4) is

$$\mathbb{E}(A^3) = \dot{\psi}_r \dot{\psi}_s \dot{\psi}_t \text{cum}(\Delta^r, \Delta^s, \Delta^t) + O(n^{-3}),$$

where $\text{cum}(\Delta^r, \Delta^s, \Delta^t)$ admits the asymptotic expansion given in [Pace and Salvan \(1997, equation \(9.74\)\)](#). For the second term,

$$\begin{aligned} E(A^2 B) &= \frac{1}{2} \dot{\psi}_r \dot{\psi}_s \ddot{\psi}_{tu} E[\Delta^r \Delta^s (\Delta^t \Delta^u - i^{tu})] + O(n^{-3}) \\ &= \frac{1}{2} \dot{\psi}_r \dot{\psi}_s \ddot{\psi}_{tu} \left\{ E(\Delta^r \Delta^s \Delta^t \Delta^u) - i^{tu} E(\Delta^r \Delta^s) \right\} + O(n^{-3}) \\ &= \dot{\psi}_r \dot{\psi}_s \ddot{\psi}_{tu} i^{rt} i^{su} + O(n^{-3}). \end{aligned} \quad (\text{S5})$$

The last equality in (S5) follows from the moment–cumulant identity (see [McCullagh, 2018, Section 3.6](#)),

$$E(\Delta^r \Delta^s \Delta^t \Delta^u) = \text{cum}(\Delta^r, \Delta^s, \Delta^t, \Delta^u) + i^{rs} i^{tu} + i^{rt} i^{su} + i^{ru} i^{st} + O(n^{-3}),$$

together with the expansion in [Pace and Salvan \(1997, equation \(9.75\)\)](#), which shows that

$$\text{cum}(\Delta^r, \Delta^s, \Delta^t, \Delta^u) = O(n^{-3}),$$

so that the terms involving $i^{rs} i^{tu}$ cancel out. Therefore, $C_3 = \kappa_3 + O(n^{-3})$, where

$$\kappa_3 = \dot{\psi}_r \dot{\psi}_s \dot{\psi}_t \text{cum}(\Delta^r, \Delta^s, \Delta^t) + 3 \dot{\psi}_r \dot{\psi}_s \ddot{\psi}_{tu} i^{rt} i^{su} \quad (\text{S6})$$

To express the first term in (S6) in matrix notation, let $s_t = \partial l(\boldsymbol{\theta}; \mathbf{Y}) / \partial \theta^t$. Then, [Pace and Salvan \(1997, equation \(9.74\)\)](#) can be written as

$$\text{cum}(\Delta^r, \Delta^s, \Delta^t) = -6 i^{rv} i^{sw} i^{tu} \left\{ \frac{1}{3} E(s_v s_w s_u) - \frac{1}{2} E(j_{wu} s_v) \right\},$$

where the error is order $O(n^{-3})$. Let $\mathbf{P}_t \equiv \mathbf{P}_t(\boldsymbol{\theta}) = E(\mathbf{s} \mathbf{s}^\top s_t)$ and $\mathbf{Q}_t \equiv \mathbf{Q}_t(\boldsymbol{\theta}) = -E(\mathbf{j} s_t)$, $\mathbf{V} = \mathbf{i}^{-1}$, and $\boldsymbol{\omega} = \mathbf{V} \dot{\boldsymbol{\psi}}$, with ω_t the t th element of $\boldsymbol{\omega}$ ($t = 1, \dots, p$). It follows that,

$$\dot{\psi}_r \dot{\psi}_s \dot{\psi}_t \text{cum}(\Delta^r, \Delta^s, \Delta^t) = -6 \sum_{t=1}^p \omega_t \boldsymbol{\omega}^\top \left(\frac{1}{3} \mathbf{P}_t + \frac{1}{2} \mathbf{Q}_t \right) \boldsymbol{\omega},$$

where the error is order $O(n^{-3})$. Moreover, for the second term in (S6), we can write

$$3 \dot{\psi}_r \dot{\psi}_s \ddot{\psi}_{tu} i^{rt} i^{su} = 3 \boldsymbol{\omega}^\top \ddot{\boldsymbol{\psi}} \boldsymbol{\omega}.$$

Substituting these two identities into (S6), we obtain

$$\kappa_3 = -6 \sum_{t=1}^p \omega_t \boldsymbol{\omega}^\top \left(\frac{1}{3} \mathbf{P}_t + \frac{1}{2} \mathbf{Q}_t \right) \boldsymbol{\omega} + 3 \boldsymbol{\omega}^\top \ddot{\boldsymbol{\psi}} \boldsymbol{\omega}. \quad (\text{S7})$$

Therefore, $K_3 = \kappa_3 + O(n^{-3})$, with κ_3 given by (S7).

S3 Example 5: Derivation of quantities required for $\tilde{\boldsymbol{\psi}}$

For the model in Example 5, let $\boldsymbol{\eta} = (\eta_1, \dots, \eta_q)^\top$, and let $\text{vec}(\boldsymbol{\Xi})$ denote the vector obtained by stacking the columns of $\boldsymbol{\Xi}$. The reference parameter is $\boldsymbol{\theta} = (\boldsymbol{\mu}^\top, \boldsymbol{\nu}^\top, \text{vec}(\boldsymbol{\Xi})^\top, \boldsymbol{\eta}^\top, \alpha, \gamma, \boldsymbol{\beta}^\top, \boldsymbol{\delta}^\top, \sigma^2)^\top$, with λ and $\mathbf{L}$ fixed. The single-observation log-likelihood contribution is

$$l(\boldsymbol{\theta}; A_i, \mathbf{x}_i, \mathbf{M}_i, Y_i) = \log f_M(\mathbf{M}_i \mid A_i, \mathbf{x}_i; \boldsymbol{\theta}) + \log f_Y(Y_i \mid \mathbf{M}_i, A_i, \mathbf{x}_i; \boldsymbol{\theta}),$$

where the two components involve separate parameter blocks of $\boldsymbol{\theta}$. Let $\mathbf{Z}$ have rows $\mathbf{z}_i^\top = (1, A_i, \mathbf{x}_i^\top)$, and let $\mathbf{C}$ be the $q \times (p+2)$ matrix whose first two columns are $\boldsymbol{\mu}$ and $\boldsymbol{\nu}$, with the remaining columns

given by Ξ . The transformation $\mathbf{W}_i = \mathbf{L}^{-1}\mathbf{M}_i$ gives $W_{ij} = \mathbf{z}_i^\top \boldsymbol{\xi}_j + e^{\eta_j} U_{ij}$, where $\boldsymbol{\xi}_j^\top$ is row j of $\mathbf{L}^{-1}\mathbf{C}$. For one transformed regression, fix j and let $\boldsymbol{\vartheta}_j = (\boldsymbol{\xi}_j^\top, \eta_j)^\top$. Let T have density $f(t; \lambda) = 2\phi(t)\Phi(\lambda t)$ and define $\mu_\lambda = \lambda(2/\pi)^{1/2}/(1 + \lambda^2)^{1/2}$, $c_\lambda = (1 - \mu_\lambda^2)^{1/2}$ and $U = (T - \mu_\lambda)/c_\lambda$. Then $E(U) = 0$ and $\text{var}(U) = 1$, and each U_{ij} has the distribution of U . With $T_{ij} = \mu_\lambda + c_\lambda e^{-\eta_j}(W_{ij} - \mathbf{z}_i^\top \boldsymbol{\xi}_j)$, the single-observation log-likelihood contribution for $\boldsymbol{\vartheta}_j$ is

$$l(\boldsymbol{\vartheta}_j; W_{ij}) = \log(2c_\lambda) - \eta_j + \log \phi(T_{ij}) + \log \Phi(\lambda T_{ij}).$$

Define

$$r_\lambda(t) = \frac{\phi(\lambda t)}{\Phi(\lambda t)}, \quad v_\lambda(t) = t - \lambda r_\lambda(t), \quad K_\lambda(t) = 1 + \lambda^2 r_\lambda(t) \{\lambda t + r_\lambda(t)\}.$$

Since $d \log\{\phi(t)\Phi(\lambda t)\}/dt = -v_\lambda(t)$ and $v'_\lambda(t) = K_\lambda(t)$, differentiation gives

$$\begin{aligned} \mathbf{s}_{\boldsymbol{\xi}_j}(\boldsymbol{\vartheta}_j; W_{ij}) &= c_\lambda e^{-\eta_j} v_\lambda(T_{ij}) \mathbf{z}_i, \\ s_{\eta_j}(\boldsymbol{\vartheta}_j; W_{ij}) &= (T_{ij} - \mu_\lambda) v_\lambda(T_{ij}) - 1, \\ \mathbf{j}_{\boldsymbol{\xi}_j \boldsymbol{\xi}_j}(\boldsymbol{\vartheta}_j; W_{ij}) &= c_\lambda^2 e^{-2\eta_j} K_\lambda(T_{ij}) \mathbf{z}_i \mathbf{z}_i^\top, \\ \mathbf{j}_{\boldsymbol{\xi}_j \eta_j}(\boldsymbol{\vartheta}_j; W_{ij}) &= c_\lambda e^{-\eta_j} \{v_\lambda(T_{ij}) + (T_{ij} - \mu_\lambda) K_\lambda(T_{ij})\} \mathbf{z}_i, \\ j_{\eta_j \eta_j}(\boldsymbol{\vartheta}_j; W_{ij}) &= (T_{ij} - \mu_\lambda) v_\lambda(T_{ij}) + (T_{ij} - \mu_\lambda)^2 K_\lambda(T_{ij}), \end{aligned}$$

where $\mathbf{s}(\boldsymbol{\vartheta}_j; W_{ij})$ is the score and $\mathbf{j}(\boldsymbol{\vartheta}_j; W_{ij})$ is the observed information. Put

$$\begin{aligned} J &= c_\lambda^2 E\{v_\lambda(T)^2\}, \\ J_{01} &= c_\lambda E[v_\lambda(T)\{(T - \mu_\lambda)v_\lambda(T) - 1\}], \\ J_{11} &= E[\{(T - \mu_\lambda)v_\lambda(T) - 1\}^2]. \end{aligned}$$

The information identity gives $J = c_\lambda^2 E\{K_\lambda(T)\}$, and the expected information is

$$\mathbf{i}(\boldsymbol{\vartheta}_j) = \begin{bmatrix} e^{-2\eta_j} J \mathbf{Z}^\top \mathbf{Z} & e^{-\eta_j} J_{01} \mathbf{Z}^\top \mathbf{1}_n \\ e^{-\eta_j} J_{01} \mathbf{1}_n^\top \mathbf{Z} & n J_{11} \end{bmatrix}.$$

Write $\mathbf{G} = (\mathbf{Z}^\top \mathbf{Z})^{-1}$ and $D_0 = J_{11} - J_{01}^2/J$. If $\mathbf{e}_0$ selects the intercept, $\mathbf{G} \mathbf{Z}^\top \mathbf{1}_n = \mathbf{e}_0$, and the blocks of $\mathbf{V}(\boldsymbol{\vartheta}_j) = \mathbf{i}(\boldsymbol{\vartheta}_j)^{-1}$ are

$$\mathbf{V}_{\boldsymbol{\xi}_j \boldsymbol{\xi}_j} = e^{2\eta_j} \left\{ \frac{\mathbf{G}}{J} + \frac{J_{01}^2 \mathbf{e}_0 \mathbf{e}_0^\top}{n J^2 D_0} \right\}, \quad \mathbf{V}_{\boldsymbol{\xi}_j \eta_j} = -\frac{e^{\eta_j} J_{01} \mathbf{e}_0}{n J D_0}, \quad V_{\eta_j \eta_j} = \frac{1}{n D_0}.$$

Let $\mathbf{e}_A$ select the treatment coefficient in $\boldsymbol{\xi}_j$, and put $g_i = \mathbf{z}_i^\top \mathbf{G} \mathbf{e}_A$, $h_i = \mathbf{z}_i^\top \mathbf{G} \mathbf{z}_i$, and $G_{AA} = \mathbf{e}_A^\top \mathbf{G} \mathbf{e}_A$. Then $\sum_i g_i = 0$, $\sum_i g_i^2 = G_{AA}$, and

$$\boldsymbol{\omega} = \mathbf{V}(\boldsymbol{\vartheta}_j) \begin{bmatrix} \mathbf{e}_A \\ 0 \end{bmatrix} = \begin{bmatrix} e^{2\eta_j} \mathbf{G} \mathbf{e}_A / J \\ 0 \end{bmatrix}.$$

Consequently,

$$\boldsymbol{\omega}^\top \mathbf{s}(\boldsymbol{\vartheta}_j; W_{ij}) = c_\lambda e^{\eta_j} v_\lambda(T_{ij}) g_i / J, \quad \boldsymbol{\omega}^\top \mathbf{j}(\boldsymbol{\vartheta}_j; W_{ij}) \boldsymbol{\omega} = c_\lambda^2 e^{2\eta_j} K_\lambda(T_{ij}) g_i^2 / J^2,$$

and the leading term of the variance of the estimator of the treatment coefficient $\mathbf{e}_A^\top \boldsymbol{\xi}_j$ is $e^{2\eta_j} G_{AA} / J$. The general expressions for the leading terms of the bias and third cumulant yield, respectively,

$$\begin{aligned} e^{\eta_j} B_A &= -\frac{1}{2} \sum_i E \left[\{\boldsymbol{\omega}^\top \mathbf{s}(\boldsymbol{\vartheta}_j; W_{ij})\} \left\{ \mathbf{s}(\boldsymbol{\vartheta}_j; W_{ij})^\top \mathbf{V}(\boldsymbol{\vartheta}_j) \mathbf{s}(\boldsymbol{\vartheta}_j; W_{ij}) - \text{tr}\{\mathbf{V}(\boldsymbol{\vartheta}_j) \mathbf{j}(\boldsymbol{\vartheta}_j; W_{ij})\} \right\} \right], \\ e^{3\eta_j} T_A &= \sum_i E \left[-2\{\boldsymbol{\omega}^\top \mathbf{s}(\boldsymbol{\vartheta}_j; W_{ij})\}^3 + 3\{\boldsymbol{\omega}^\top \mathbf{j}(\boldsymbol{\vartheta}_j; W_{ij}) \boldsymbol{\omega}\} \{\boldsymbol{\omega}^\top \mathbf{s}(\boldsymbol{\vartheta}_j; W_{ij})\} \right]. \end{aligned}$$

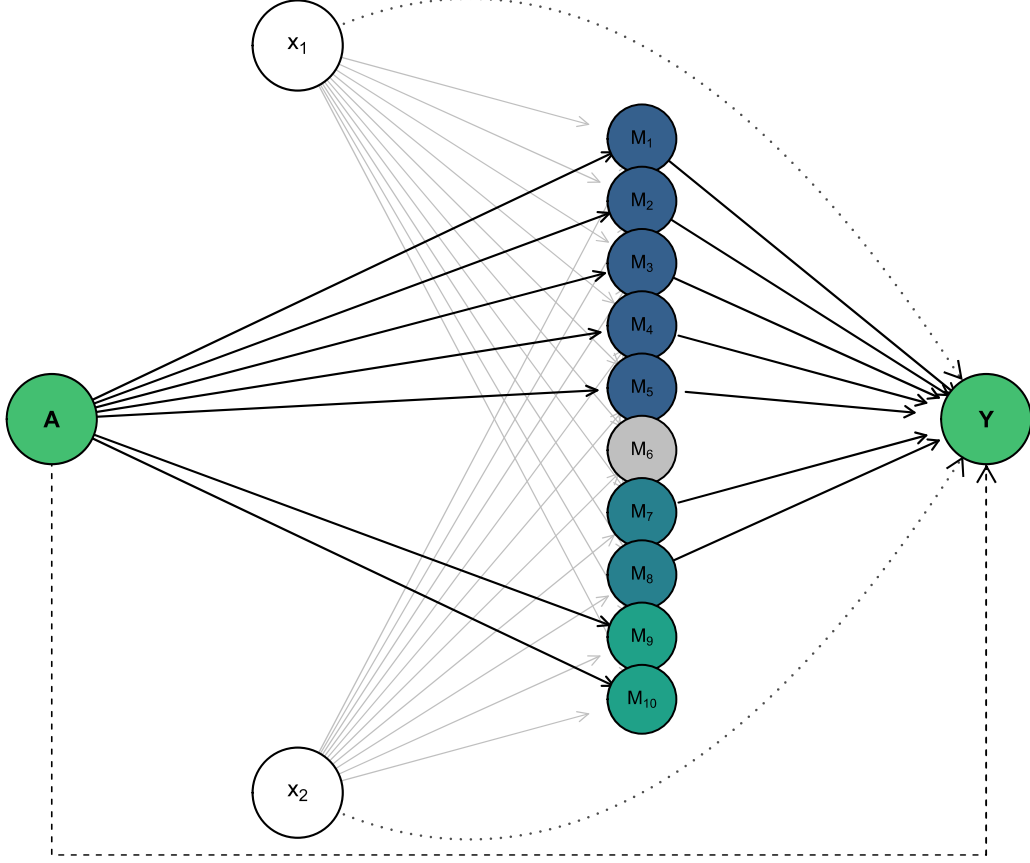


Figure S1: Causal diagram for model (17) with $q = 10$. Treatment A and baseline covariates $\mathbf{x} = (x_1, x_2)^\top$ affect the mediators $M_1, \dots, M_{10}$, and A , $\mathbf{x}$ and the mediators affect the outcome Y . Treatment is randomized independently of $\mathbf{x}$, so that there is no edge between A and $\mathbf{x}$. The edges $A \rightarrow M_j$ and $M_j \rightarrow Y$ correspond to non-zero values of the coefficients ν_j and β_j , respectively, in the simulation design described in Example 5. Thus, $\nu_j \beta_j \neq 0$ for $j = 1, \dots, 5$, $\nu_6 = \beta_6 = 0$, $\nu_j = 0$ and $\beta_j \neq 0$ for $j = 7, 8$, and $\nu_j \neq 0$ and $\beta_j = 0$ for $j = 9, 10$.

Substitution of the information matrix blocks yields $B_A = C_B \sum_i g_i h_i$ and $T_A = C_3 \sum_i g_i^3$, where

$$C_B = -c_\lambda^3 \mathbb{E}\{v_\lambda(T)^3 - v_\lambda(T)K_\lambda(T)\}/2J^2, \quad C_3 = c_\lambda^3 \mathbb{E}\{-2v_\lambda(T)^3 + 3v_\lambda(T)K_\lambda(T)\}/J^3.$$

The additional terms in the expression for the leading term of the bias cancel after taking expectations and summing, since they are proportional to $\sum_i g_i = 0$. The constants J, C_B, C_3 depend only on λ and are computed by one-dimensional numerical integration.

Returning to the reference parameter $\boldsymbol{\theta}$, let $\boldsymbol{\nu}^*$ contain the transformed treatment coefficients. Since $\boldsymbol{\nu} = \mathbf{L}\boldsymbol{\nu}^*$, conditional independence of the transformed regressions gives $\mathbf{V}_\nu = \mathbf{L}(G_{AA}\mathbf{D}^2/J)\mathbf{L}^\top = G_{AA}\boldsymbol{\Sigma}_M/J$ as the leading term of the covariance matrix of $\hat{\boldsymbol{\nu}}$. Writing $\boldsymbol{\rho} = \mathbf{D}\mathbf{L}^\top\boldsymbol{\beta}$, the corresponding mean-bias and third-cumulant contributions to $\hat{\psi}$ are $B_A \sum_j \rho_j$ and $T_A \sum_j \rho_j^3$. For the Gaussian outcome regression, the Schur complement of the nuisance regression block in expected information is $n\boldsymbol{\Sigma}_M/\sigma^2$, giving $\mathbf{V}_\beta = \sigma^2\boldsymbol{\Sigma}_M^{-1}/n$ as the leading term of the covariance matrix of $\hat{\boldsymbol{\beta}}$. The identity $\hat{\psi} - \psi = \boldsymbol{\beta}^\top(\hat{\boldsymbol{\nu}} - \boldsymbol{\nu}) + \boldsymbol{\nu}^\top(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}) + (\hat{\boldsymbol{\nu}} - \boldsymbol{\nu})^\top(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta})$ and conditional unbiasedness of $\hat{\boldsymbol{\beta}}$ imply that only the first term contributes to the mean bias of $\hat{\psi}$. Thus, the leading term of the mean bias is $b_{(\psi)} = B_A \sum_{j=1}^q \rho_j$. The block-diagonal expected information implies that the leading term of the variance of $\hat{\psi}$ is $\kappa_2 = \boldsymbol{\beta}^\top \mathbf{V}_\nu \boldsymbol{\beta} + \boldsymbol{\nu}^\top \mathbf{V}_\beta \boldsymbol{\nu}$.

Since $\psi = h(\boldsymbol{\theta}) = \boldsymbol{\nu}^\top \boldsymbol{\beta}$, the only nonzero blocks of $\dot{\boldsymbol{\psi}}$ are $\dot{\boldsymbol{\psi}}_{(\boldsymbol{\nu})} = \boldsymbol{\beta}$ and $\dot{\boldsymbol{\psi}}_{(\boldsymbol{\beta})} = \boldsymbol{\nu}$. The only nonzero blocks of $\ddot{\boldsymbol{\psi}}$ are the two mixed blocks corresponding to $\boldsymbol{\nu}$ and $\boldsymbol{\beta}$, both equal to $\mathbf{I}_q$. The last

Table S1: Comparison of the ML focus estimator ($\hat{\psi}$) and the MBC focus estimator ($\tilde{\psi}$) of the joint indirect effect for $\tau = 1$ and $n_1 = n/8$, in terms of simulation-based estimates (see Example 5) of probability of underestimation (PU), mean bias (BIAS), and root mean squared error (RMSE). The column “Wald” reports the coverage of nominally 95% Wald-type intervals using the estimated standard error evaluated at the supplied maximum likelihood estimate of θ . The column “HulC” reports the coverage of nominally 95% HulC-type intervals with $\Delta = 0$, constructed using the corresponding estimator. All summaries are $\times 100$.

q	Estimator	n	PU	BIAS	RMSE	Wald	HulC
10	$\hat{\psi}$	256	44.0	0.4	2.1	97.7	93.6
		512	45.4	0.2	1.3	97.0	93.6
		1024	46.8	0.1	0.8	96.2	94.0
	$\tilde{\psi}$	256	50.1	0.1	2.1	97.7	95.0
		512	50.0	0.0	1.3	96.9	95.0
		1024	50.2	0.0	0.8	96.1	95.0
20	$\hat{\psi}$	256	44.2	0.4	2.6	97.9	94.3
		512	45.1	0.2	1.4	97.7	94.1
		1024	46.5	0.1	0.9	97.0	94.1
	$\tilde{\psi}$	256	49.9	0.1	2.5	98.2	95.0
		512	50.0	0.0	1.4	97.8	94.9
		1024	50.3	0.0	0.9	97.0	95.0
30	$\hat{\psi}$	256	44.8	0.4	3.0	97.7	94.7
		512	45.3	0.2	1.7	97.7	94.5
		1024	46.6	0.1	1.0	97.1	94.4
	$\tilde{\psi}$	256	50.0	0.1	2.9	98.2	95.0
		512	49.9	0.0	1.6	98.0	95.1
		1024	50.3	0.0	1.0	97.3	95.2

term in the general expression for the leading term of the third cumulant of $\hat{\psi}$ reduces to

$$3 \begin{bmatrix} V_\nu \beta \\ V_\beta \nu \end{bmatrix}^\top \begin{bmatrix} \mathbf{0} & I_q \\ I_q & \mathbf{0} \end{bmatrix} \begin{bmatrix} V_\nu \beta \\ V_\beta \nu \end{bmatrix} = 6\beta^\top V_\nu V_\beta \nu.$$

The remaining contributions cancel under the model assumptions and the identity $\sum_i g_i = 0$. Combining these expressions gives $\kappa_3 = T_A \sum_{j=1}^q \rho_j^3 + 6\beta^\top V_\nu V_\beta \nu$. At $\lambda = 0$, $J = 1$ and $C_B = C_3 = 0$. Evaluating $b_{(\psi)}$, κ_2 and κ_3 at the ML estimator $\hat{\theta}$ and using expression (4) yields the median bias-corrected estimator $\tilde{\psi}$.

S4 Example 5: Supplementary results

Following the simulation design described in Example 5, Figure S1 presents the causal diagram for $q = 10$.

Additional simulation results for $\tau = 1, 4$ are reported in Tables S1 and S2 for $n_1 = n/8$, and Tables S3 and S4 for $n_1 = 3n/32$, respectively. Table S5 reports results for $\tau = 4$ and $n_1 = n/16$.

Table S2: Comparison of the ML focus estimator ($\hat{\psi}$) and the MBC focus estimator ($\tilde{\psi}$) of the joint indirect effect for $\tau = 4$ and $n_1 = n/8$, in terms of simulation-based estimates (see Example 5) of probability of underestimation (PU), mean bias (BIAS), and root mean squared error (RMSE). The column “Wald” reports the coverage of nominally 95% Wald-type intervals using the estimated standard error evaluated at the supplied maximum likelihood estimate of θ . The column “HulC” reports the coverage of nominally 95% HulC-type intervals with $\Delta = 0$, constructed using the corresponding estimator. All summaries are $\times 100$.

q	Estimator	n	PU	BIAS	RMSE	Wald	HulC
10	$\hat{\psi}$	256	38.8	1.6	5.5	93.8	83.6
		512	41.7	0.8	3.6	94.4	87.8
		1024	44.0	0.4	2.5	94.8	91.2
	$\tilde{\psi}$	256	49.4	0.2	5.2	94.6	94.5
		512	49.8	0.1	3.5	94.8	94.8
		1024	49.8	0.0	2.5	94.9	94.9
20	$\hat{\psi}$	256	35.4	1.7	4.7	94.6	85.7
		512	38.9	0.9	3.0	94.7	86.0
		1024	42.1	0.4	2.0	95.0	88.9
	$\tilde{\psi}$	256	49.4	0.2	4.4	95.8	94.5
		512	49.7	0.1	2.9	95.6	94.8
		1024	50.1	0.0	1.9	95.3	94.9
30	$\hat{\psi}$	256	34.9	1.7	4.6	95.1	90.6
		512	38.2	0.8	2.8	95.4	87.3
		1024	40.9	0.4	1.8	95.3	88.4
	$\tilde{\psi}$	256	49.5	0.2	4.2	96.6	94.9
		512	50.0	0.1	2.6	96.1	94.9
		1024	49.9	0.0	1.7	95.8	95.0

S5 Example 3 continued: Individual marginal effects for generalized linear models

Popular and well-used frameworks for marginal effects like the `marginalEffects` R package rely on Wald-type procedures for inference. Continuing from Example 3, Figure S2 shows the coverage of 95% Wald-type confidence intervals for the individual marginal effects, using both supplied and compatible (see Section 4.1) estimates for the computation of the standard errors, and HulC-type 95% confidence intervals. The Wald-type confidence intervals using the MBC estimator show coverage closer to the nominal level than both the mean bias-corrected estimator and the ML estimator, but still undercover. Basing the standard error computation on the compatible estimates of θ brings the performance of intervals based on reduced mean-bias estimators of Firth (1993) and those based on the ML estimator closer to each other than intervals based on supplied values. The above experimental results demonstrate the difficulties with Wald-type inference for marginal effects, even in situations with seemingly abundant information about the parameters of generalized linear models (2500 observations for 6 parameters in this example). Although all estimators are asymptotically median unbiased, as expected, the MBC estimators are the only ones for which HulC intervals with $\Delta = 0$ achieve near-nominal finite-sample coverage.

Table S3: Comparison of the ML focus estimator ($\hat{\psi}$) and the MBC focus estimator ($\tilde{\psi}$) of the joint indirect effect for $\tau = 1$ and $n_1 = 3n/32$, in terms of simulation-based estimates (see Example 5) of probability of underestimation (PU), mean bias (BIAS), and root mean squared error (RMSE). The column “Wald” reports the coverage of nominally 95% Wald-type intervals using the estimated standard error evaluated at the supplied maximum likelihood estimate of θ . The column “HulC” reports the coverage of nominally 95% HulC-type intervals with $\Delta = 0$, constructed using the corresponding estimator. All summaries are $\times 100$.

q	Estimator	n	PU	BIAS	RMSE	Wald	HulC
10	$\hat{\psi}$	256	42.2	0.6	2.4	97.7	92.6
		512	43.7	0.3	1.4	97.2	92.8
		1024	45.5	0.1	0.9	96.3	93.1
	$\tilde{\psi}$	256	49.9	0.1	2.3	97.9	94.9
		512	50.0	0.1	1.4	97.1	95.1
		1024	50.1	0.0	0.9	96.3	94.8
20	$\hat{\psi}$	256	42.4	0.6	2.9	97.9	93.7
		512	43.6	0.3	1.6	97.8	93.4
		1024	45.0	0.1	1.0	97.1	93.5
	$\tilde{\psi}$	256	49.8	0.1	2.7	98.4	94.9
		512	50.0	0.0	1.5	98.0	94.9
		1024	50.1	0.0	0.9	97.3	95.1
30	$\hat{\psi}$	256	43.1	0.6	3.4	97.7	94.7
		512	43.8	0.3	1.8	97.8	94.0
		1024	45.4	0.1	1.1	97.3	93.8
	$\tilde{\psi}$	256	49.8	0.1	3.2	98.3	95.1
		512	49.9	0.0	1.7	98.3	95.0
		1024	50.3	0.0	1.0	97.6	94.9

S6 Example 6: Derivation of quantities required for $\tilde{\psi}$

Let $Y_1, \dots, Y_n$ be independent and identically distributed according to a Weibull distribution with scale parameter $e^\eta > 0$ and shape parameter $\tau > 0$. Suppose that the focus parameter is the $(1 - \alpha)$ quantile of the Weibull distribution $\psi = \exp(\eta + c_\alpha/\tau)$, where $\alpha \in (0, 1)$ and $c_\alpha = \log(-\log \alpha)$. Differentiating the focus parameter with respect to the reference parameters $\theta = (\eta, \tau)^\top$ gives

$$\dot{\psi} = \psi \begin{bmatrix} 1 \\ -c_\alpha/\tau^2 \end{bmatrix}, \quad \ddot{\psi} = \psi \begin{bmatrix} 1 & -c_\alpha/\tau^2 \\ -c_\alpha/\tau^2 & 2c_\alpha/\tau^3 + c_\alpha^2/\tau^4 \end{bmatrix}.$$

For a single realization Y_i , the log-likelihood is

$$l(\theta; Y_i) = \log \tau - \tau \eta + (\tau - 1) \log Y_i - \exp\{\tau(\log Y_i - \eta)\}.$$

Define

$$T_i = \log Y_i - \eta, \quad W_i = \exp(\tau T_i) = (Y_i/e^\eta)^\tau.$$

It follows that $W_i \sim \text{Exp}(1)$ and $T_i = L_i/\tau$, where $L_i = \log W_i$. Set $d_\alpha = c_\alpha + \gamma - 1$. Let γ denote the Euler-Mascheroni constant, ζ_3 the Riemann zeta function at 3, and $\Gamma(\cdot)$ the gamma function.

Table S4: Comparison of the ML focus estimator ($\hat{\psi}$) and the MBC focus estimator ($\tilde{\psi}$) of the joint indirect effect for $\tau = 4$ and $n_1 = 3n/32$, in terms of simulation-based estimates (see Example 5) of probability of underestimation (PU), mean bias (BIAS), and root mean squared error (RMSE). The column “Wald” reports the coverage of nominally 95% Wald-type intervals using the estimated standard error evaluated at the supplied maximum likelihood estimate of θ . The column “HulC” reports the coverage of nominally 95% HulC-type intervals with $\Delta = 0$, constructed using the corresponding estimator. All summaries are $\times 100$.

q	Estimator	n	PU	BIAS	RMSE	Wald	HulC
10	$\hat{\psi}$	256	35.8	2.3	6.4	93.0	75.6
		512	39.7	1.1	4.2	94.0	83.5
		1024	42.6	0.6	2.8	94.5	88.6
	$\tilde{\psi}$	256	49.3	0.3	6.0	94.5	93.3
		512	49.8	0.1	4.0	94.6	94.7
		1024	49.9	0.0	2.8	94.8	94.9
20	$\hat{\psi}$	256	31.8	2.4	5.6	93.5	79.6
		512	36.1	1.2	3.4	94.2	80.1
		1024	40.2	0.6	2.3	94.7	85.1
	$\tilde{\psi}$	256	49.0	0.3	4.9	95.7	93.7
		512	49.5	0.1	3.2	95.5	94.7
		1024	50.0	0.0	2.2	95.3	95.0
30	$\hat{\psi}$	256	30.9	2.4	5.4	94.4	87.4
		512	34.8	1.2	3.2	94.9	82.4
		1024	38.4	0.6	2.0	95.0	84.2
	$\tilde{\psi}$	256	49.2	0.2	4.7	96.6	94.5
		512	49.7	0.1	2.9	96.1	94.7
		1024	49.7	0.0	1.9	95.7	95.0

In what follows, we repeatedly use the identities

$$E(W_i^m L_i^k) = \frac{d^k}{da^k} \Gamma(a) \Big|_{a=m+1},$$

so that in particular, $E(W_i) = 1$, $E(W_i^2) = 2$, $E(W_i^3) = 6$, $E(L_i) = -\gamma$, $E(L_i^2) = \pi^2/6 + \gamma^2$, $E(L_i^3) = -2\zeta_3 - \gamma\pi^2/2 - \gamma^3$, and

$$\begin{aligned} E(W_i L_i) &= 1 - \gamma, & E(W_i^2 L_i) &= 3 - 2\gamma, & E(W_i^3 L_i) &= 11 - 6\gamma, \\ E(W_i L_i^2) &= \pi^2/6 + \gamma^2 - 2\gamma, & E(W_i L_i^3) &= 2 - 2\zeta_3 - (6 - \pi^2)(1 - \gamma)/2 + (1 - \gamma)^3, \\ E(W_i^2 L_i^2) &= 2\gamma^2 - 6\gamma + 2 + \pi^2/3, & E(W_i^3 L_i^2) &= 6\gamma^2 - 22\gamma + 12 + \pi^2, \\ E(W_i^2 L_i^3) &= -2\gamma^3 + 9\gamma^2 - \pi^2\gamma - 6\gamma - 4\zeta_3 + 3\pi^2/2, \\ E(W_i^3 L_i^3) &= -6\gamma^3 + 33\gamma^2 - 36\gamma - 3\pi^2\gamma - 12\zeta_3 + 6 + 11\pi^2/2. \end{aligned}$$

Per observation, the score components are

$$s_\eta(\theta; Y_i) = \tau(W_i - 1), \quad s_\tau(\theta; Y_i) = (1 + L_i\{1 - W_i\})/\tau.$$

Table S5: Comparison of the ML focus estimator ($\hat{\psi}$) and the MBC focus estimator ($\tilde{\psi}$) of the joint indirect effect for $\tau = 4$ and $n_1 = n/16$, in terms of simulation-based estimates (see Example 5) of probability of underestimation (PU), mean bias (BIAS), and root mean squared error (RMSE). The column “Wald” reports the coverage of nominally 95% Wald-type intervals using the estimated standard error evaluated at the supplied maximum likelihood estimate of θ . The column “HulC” reports the coverage of nominally 95% HulC-type intervals with $\Delta = 0$, constructed using the corresponding estimator. All summaries are $\times 100$.

q	Estimator	n	PU	BIAS	RMSE	Wald	HulC
10	$\hat{\psi}$	256	31.7	3.6	8.1	91.2	59.6
		512	36.7	1.8	5.2	93.1	72.7
		1024	40.5	0.9	3.5	94.1	83.9
	$\tilde{\psi}$	256	48.5	0.5	7.3	94.2	90.3
		512	49.5	0.2	4.9	94.6	93.7
		1024	49.8	0.1	3.3	94.8	95.0
	$\hat{\psi}$	256	26.3	3.9	7.2	91.2	67.8
		512	31.8	1.9	4.3	93.1	67.1
		1024	36.8	0.9	2.8	93.9	76.9
20	$\tilde{\psi}$	256	47.7	0.5	6.0	95.6	91.1
		512	49.0	0.2	3.9	95.4	93.5
		1024	49.7	0.1	2.6	95.2	94.7
	$\hat{\psi}$	256	25.4	3.9	7.0	92.2	81.7
		512	30.0	1.9	4.0	93.5	71.5
		1024	34.6	0.9	2.5	94.2	75.8
	$\tilde{\psi}$	256	48.3	0.4	5.7	96.5	93.1
		512	49.1	0.2	3.5	96.1	93.7
		1024	49.7	0.1	2.3	95.7	94.7

The entries of the observed information matrix $\mathbf{j}(\theta; Y_i) = -\nabla \nabla^\top l(\theta; Y_i)$ are

$$\mathbf{j}_{\eta\eta}(\theta; Y_i) = \tau^2 W_i, \quad \mathbf{j}_{\eta\tau}(\theta; Y_i) = 1 - W_i(1 + L_i), \quad \mathbf{j}_{\tau\tau}(\theta; Y_i) = (1 + W_i L_i^2)/\tau^2.$$

Thus, the Fisher information matrix for the full sample $\mathbf{i}(\theta) = n \mathbb{E}(\mathbf{j}(\theta; Y_i))$ has entries

$$i_{\eta\eta}(\theta) = n\tau^2, \quad i_{\eta\tau}(\theta) = n(\gamma - 1), \quad i_{\tau\tau}(\theta) = nA/\tau^2,$$

where $A = \pi^2/6 + (1 - \gamma)^2$. Hence

$$\mathbf{V} = \mathbf{i}^{-1}(\theta) = \frac{6}{n\pi^2} \begin{bmatrix} A/\tau^2 & 1 - \gamma \\ 1 - \gamma & \tau^2 \end{bmatrix},$$

and $\omega = \mathbf{V}\psi$ can be expressed as

$$\omega = \frac{6\psi}{n\pi^2} \begin{bmatrix} \{\pi^2/6 - (1 - \gamma)d_\alpha\}/\tau^2 \\ -d_\alpha \end{bmatrix}.$$

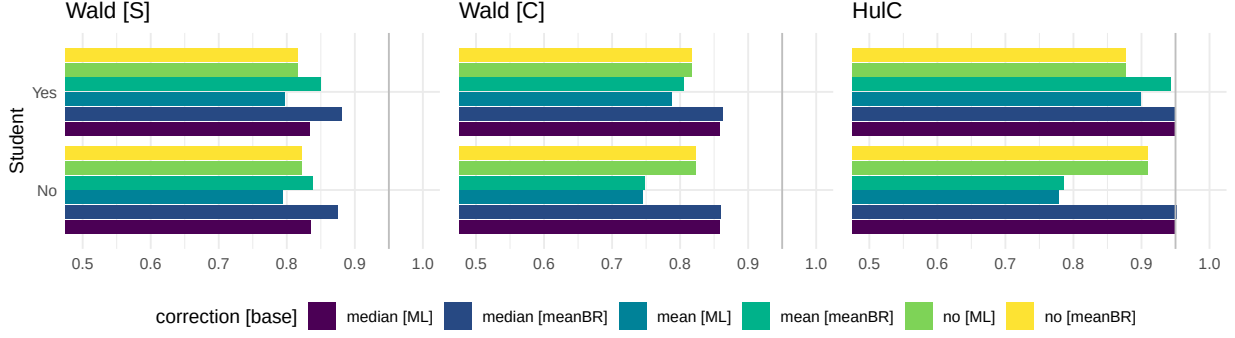


Figure S2: Comparison of various estimators of the individual marginal effects at covariate settings with **Student** = No and **Student** = Yes in Example 3. The panels report simulation-based estimates of the coverage of nominally 95% Wald-type confidence intervals based on supplied and compatible estimates of θ (“Wald [S]” and “Wald [C]”, respectively), and HulC-type confidence intervals with $\Delta = 0$. Results are shown for the median (“median”) and mean (“mean”) bias-corrected focus estimators, and for the uncorrected plug-in estimator (“no”). The label in brackets indicates whether the plug-in quantities are evaluated at ML (ML) or reduced mean-bias (meanBR) estimators of β (Firth, 1993).

The entries of the matrices $\mathbf{P}_t \equiv \mathbf{P}_t(\theta) = E(\mathbf{s}\mathbf{s}^\top s_t)$ can be computed as

$$\begin{aligned} [\mathbf{P}_\eta]_{1,1} &= n E \{ \tau^3 (W_i - 1)^3 \} = 2n\tau^3, \\ [\mathbf{P}_\eta]_{1,2} &= n E \{ \tau (W_i - 1)^2 (1 + L_i - L_i W_i) \} = 2n\tau(\gamma - 2), \\ [\mathbf{P}_\eta]_{2,2} &= n E \{ (W_i - 1)(1 + L_i - L_i W_i)^2 / \tau \} = 2nD/\tau, \\ [\mathbf{P}_\tau]_{1,1} &= n E \{ \tau (W_i - 1)^2 (1 + L_i - L_i W_i) \} = 2n\tau(\gamma - 2), \\ [\mathbf{P}_\tau]_{1,2} &= n E \{ (W_i - 1)(1 + L_i - L_i W_i)^2 / \tau \} = 2nD/\tau, \\ [\mathbf{P}_\tau]_{2,2} &= n E \{ (1 + L_i - L_i W_i)^3 / \tau^3 \} = nE/\tau^3, \end{aligned}$$

where $D = \gamma^2 - 4\gamma + 2 + \pi^2/6$ and $E = 2\gamma^3 - 12\gamma^2 + \pi^2\gamma + 12\gamma - 2\pi^2 - 2 + 4\zeta_3$. Similarly, the entries of the matrices $\mathbf{Q}_t \equiv \mathbf{Q}_t(\theta) = -E(\mathbf{j}s_t)$ are

$$\begin{aligned} [\mathbf{Q}_\eta]_{1,1} &= -n\tau^3, & [\mathbf{Q}_\tau]_{1,1} &= n\tau(1 - \gamma), \\ [\mathbf{Q}_\eta]_{1,2} &= n\tau(3 - \gamma), & [\mathbf{Q}_\tau]_{1,2} &= -nD/\tau, \\ [\mathbf{Q}_\eta]_{2,2} &= -nD/\tau, & [\mathbf{Q}_\tau]_{2,2} &= nH/\tau^3, \end{aligned}$$

where $H = -\gamma^3 + 5\gamma^2 - \gamma\pi^2/2 - 4\gamma - 2\zeta_3 + 5\pi^2/6$. Then

$$\mathbf{P}_\eta = \begin{bmatrix} 2n\tau^3 & 2n\tau(\gamma - 2) \\ 2n\tau(\gamma - 2) & 2nD/\tau \end{bmatrix}, \quad \mathbf{P}_\tau = \begin{bmatrix} 2n\tau(\gamma - 2) & 2nD/\tau \\ 2nD/\tau & nE/\tau^3 \end{bmatrix},$$

and

$$\mathbf{Q}_\eta = \begin{bmatrix} -n\tau^3 & n\tau(3 - \gamma) \\ n\tau(3 - \gamma) & -nD/\tau \end{bmatrix}, \quad \mathbf{Q}_\tau = \begin{bmatrix} n\tau(1 - \gamma) & -nD/\tau \\ -nD/\tau & nH/\tau^3 \end{bmatrix}.$$

Combining the above ingredients in equations (6), (7), and (8), and performing some straightforward algebra, yields the expressions given in (19). Substituting these expressions into formula (9) leads to the closed-form expression for the median bias-corrected estimator of the quantile parameter presented in (20).

Table S6: Comparison of the ML estimator ($\hat{\psi}$), the MBC focus estimator ($\tilde{\psi}$), and the r^* -based estimator (ψ^*) of the 0.99 quantile in terms of simulation-based estimates (see Example 6) of probability of underestimation (PU), root mean squared error (RMSE), and coverage of nominally 95% r^* -based confidence intervals, HulC-type confidence intervals with $\Delta = 0$, and Wald-type confidence intervals, based on supplied and compatible estimates of θ (“Wald [S]” and “Wald [C]”, respectively). All summaries are $\times 100$.

n	Estimator	PU	RMSE	r^*	HulC	Wald [S]	Wald [C]
25	$\hat{\psi}$	58.3	97.5		85.0	88.2	88.2
	$\tilde{\psi}$	50.3	102.7		94.0	90.2	91.4
	ψ^*	49.7	103.6	95.0			
50	$\hat{\psi}$	55.9	69.1		90.3	91.5	91.5
	$\tilde{\psi}$	49.9	71.0		94.8	92.6	93.3
	ψ^*	49.6	71.2	94.9			
100	$\hat{\psi}$	54.6	48.8		92.7	93.1	93.1
	$\tilde{\psi}$	50.3	49.4		95.0	93.7	94.1
	ψ^*	50.1	49.4	94.9			
200	$\hat{\psi}$	53.1	34.5		93.8	94.1	94.1
	$\tilde{\psi}$	50.1	34.7		94.9	94.4	94.6
	ψ^*	50.0	34.7	94.9			

S7 Example 6: Supplementary results

Table S6 and Table S7 report the results of the simulation experiment of Example 6, about estimation and inference for quantiles of the Weibull distribution, for $\alpha = 0.01$ and $\alpha = 0.10$, respectively.

S8 Example: Mahalanobis distance between two distributions

Suppose that $\mathbf{Y}_{1,1}, \dots, \mathbf{Y}_{1,n_1}$ and $\mathbf{Y}_{2,1}, \dots, \mathbf{Y}_{2,n_2}$ are independent random vectors with $\mathbf{Y}_{1,i} \sim N_p(\boldsymbol{\mu}_1, \boldsymbol{\Sigma})$ and $\mathbf{Y}_{2,j} \sim N_p(\boldsymbol{\mu}_2, \boldsymbol{\Sigma})$, where $\boldsymbol{\mu}_1, \boldsymbol{\mu}_2 \in \mathbb{R}^p$ and $\boldsymbol{\Sigma}$ is a $p \times p$ positive definite matrix. Let $n = n_1 + n_2$. The squared Mahalanobis distance between the two distributions is $\psi = (\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)^\top \boldsymbol{\Sigma}^{-1} (\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)$. Let $\bar{\mathbf{Y}}_1$ and $\bar{\mathbf{Y}}_2$ be the sample means, and define $\mathbf{W} = \sum_{i=1}^{n_1} (\mathbf{Y}_{1,i} - \bar{\mathbf{Y}}_1)(\mathbf{Y}_{1,i} - \bar{\mathbf{Y}}_1)^\top + \sum_{j=1}^{n_2} (\mathbf{Y}_{2,j} - \bar{\mathbf{Y}}_2)(\mathbf{Y}_{2,j} - \bar{\mathbf{Y}}_2)^\top$. The maximum likelihood estimator of $\boldsymbol{\Sigma}$ is $\hat{\boldsymbol{\Sigma}} = \mathbf{W}/n$, and the maximum likelihood estimator of ψ is $\hat{\psi} = (\bar{\mathbf{Y}}_1 - \bar{\mathbf{Y}}_2)^\top \hat{\boldsymbol{\Sigma}}^{-1} (\bar{\mathbf{Y}}_1 - \bar{\mathbf{Y}}_2)$. Under the model, $\mathbf{W} \sim \text{Wishart}_p(n-2, \boldsymbol{\Sigma})$ and is independent of $\bar{\mathbf{Y}}_1 - \bar{\mathbf{Y}}_2$. Hence, we can write $\hat{\psi} = cX_1/X_2$, where $c = n^2/(n_1n_2)$, X_1 is independent of X_2 , with $X_1 \sim \chi_p^2(\lambda)$ and $X_2 \sim \chi_\nu^2$, where $\lambda = n_1n_2\psi/n$, $\nu = n - p - 1$. The symbol $\chi_p^2(\lambda)$ denotes the noncentral chi-squared distribution with non-centrality parameter λ and p degrees of freedom, while χ_ν^2 is the chi-squared distribution with ν degrees of freedom. For $\nu_k = n - p - 1 - k$ and $\nu_6 > 0$, straightforward algebra gives

$$B_{(\psi)} = c \frac{m_1}{\nu_2} - \psi, \quad K_2 = c^2 \left\{ \frac{m_2}{\nu_2\nu_4} - \frac{m_1^2}{\nu_2^2} \right\}, \quad K_3 = c^3 \left\{ \frac{m_3}{\nu_2\nu_4\nu_6} - 3 \frac{m_1m_2}{\nu_2^2\nu_4} + 2 \frac{m_1^3}{\nu_2^3} \right\}, \quad (\text{S8})$$

where $m_1 = p + \lambda$, $m_2 = m_1^2 + 2(p + 2\lambda)$, and $m_3 = m_1^3 + 6m_1(p + 2\lambda) + 8(p + 3\lambda)$. All quantities in (S8) depend on $\boldsymbol{\mu}_1, \boldsymbol{\mu}_2$ and $\boldsymbol{\Sigma}$ only through ψ . Evaluating those at $\hat{\psi}$, expression (4) results in a median bias-corrected estimator of the squared Mahalanobis distance between the two distributions.

Table S7: Comparison of the ML estimator ($\hat{\psi}$), the MBC focus estimator ($\tilde{\psi}$), and the r^* -based estimator (ψ^*) of the 0.90 quantile in terms of simulation-based estimates (see Example 6) of probability of underestimation (PU), root mean squared error (RMSE), and coverage of nominally 95% r^* -based confidence intervals, HulC-type confidence intervals with $\Delta = 0$, and Wald-type confidence intervals, based on supplied and compatible estimates of θ (“Wald [S]” and “Wald [C]”, respectively). All summaries are $\times 100$.

n	Estimator	PU	RMSE	r^*	HulC	Wald [S]	Wald [C]
25	$\hat{\psi}$	56.6	48.8		89.6	90.4	90.4
	$\tilde{\psi}$	50.4	49.9		94.7	91.4	92.6
	ψ^*	49.9	50.1	95.0			
50	$\hat{\psi}$	54.5	34.7		92.1	92.4	92.4
	$\tilde{\psi}$	50.3	35.1		95.0	93.0	93.7
	ψ^*	50.1	35.1	95.0			
100	$\hat{\psi}$	53.5	24.4		93.8	93.7	93.7
	$\tilde{\psi}$	50.2	24.5		95.1	94.1	94.4
	ψ^*	50.2	24.5	95.0			
200	$\hat{\psi}$	52.0	17.2		94.3	94.4	94.4
	$\tilde{\psi}$	49.7	17.3		94.9	94.5	94.7
	ψ^*	49.7	17.3	95.0			

Table S8 compares the estimation performance of $\hat{\psi}$ and $\tilde{\psi}$, and of Wald-type confidence intervals based on them, by simulating 1000000 samples for each combination of $n = 2^q$, $q \in \{7, 8, 9, 10\}$, and $p \in \{10, 20, 30\}$ from the two-sample normal model with $n_1 = n_2 = n/2$, $\mu_1 = \mathbf{0}_p$, $\mu_2 = (1/p, 2/p, \dots, 1)^\top$ and $\Sigma_{ij} = 1/2^{|i-j|}$. The median bias-corrected estimator has percentage of underestimation substantially closer to 50% than the maximum likelihood estimator in all cases, and delivers improvements in terms of finite-sample mean bias, mean absolute deviation and root mean squared error. Furthermore, naive Wald-type confidence intervals based on $\tilde{\psi}$ are generally closer to the 95% nominal level, especially for large p .

S9 Example: Circular variance in the bivariate von Mises sine model

Suppose that $\mathbf{Y}_1, \dots, \mathbf{Y}_n$, with $\mathbf{Y}_i = (Y_{1i}, Y_{2i})^\top$, are independent bivariate angular random vectors taking values in $[0, 2\pi)^2$, following the bivariate von Mises sine distribution (Singh *et al.*, 2002; see also Mardia and Frellsen, 2012) with density

$$f(\mathbf{y}; \theta) = C(\theta)^{-1} \exp \{ e^{\tau_1} \cos(y_1 - \mu_1) + e^{\tau_2} \cos(y_2 - \mu_2) + \lambda \sin(y_1 - \mu_1) \sin(y_2 - \mu_2) \},$$

where $\mathbf{y} = (y_1, y_2)^\top$, $\theta = (\mu_1, \mu_2, \tau_1, \tau_2, \lambda)^\top$, $\mu_1, \mu_2 \in [0, 2\pi)$, $\tau_1, \tau_2, \lambda \in \mathbb{R}$. We consider as focus parameter the circular variance of the first angular component, $\psi = h(\theta) = 1 - \mathbb{E}\{\cos(Y_{1i} - \mu_1)\}$, which takes values in $(0, 1)$. This is a smooth function of θ and can be evaluated numerically, for instance using the function `circ_varcor_model()` from the BAMB I R package (Chakraborty and Wong, 2021). Similarly, the maximum likelihood estimator $\hat{\theta}$ can be obtained numerically using the function `vm2_mle()` from the same BAMB I package, which evaluates the likelihood through a numerical approximation of the normalizing constant $C(\theta)$ based on its infinite-series representation. The plug-in maximum likelihood estimator of ψ is then $\hat{\psi} = h(\hat{\theta})$.

Table S8: Comparison of the ML estimator ($\hat{\psi}$) and the MBC focus estimator ($\tilde{\psi}$) of the squared Mahalanobis distance between two multivariate normal distributions in terms of simulation-based estimates of mean bias (BIAS), mean absolute deviation (MAD), probability of underestimation (PU), root mean squared error (RMSE), and coverage of 95% Wald-type confidence intervals, where the standard error is estimated by $\{K_2(\hat{\psi})\}^{1/2}$ and $\{K_2(\tilde{\psi})\}^{1/2}$, respectively. All summaries are $\times 100$.

p	Estimator	n	BIAS	MAD	PU	RMSE	Coverage
10	$\hat{\psi}$	128	53.8	63.3	19.1	81.6	97.1
		256	25.5	36.4	27.0	46.8	95.9
		512	12.4	22.9	33.3	29.2	95.3
		1024	6.1	15.2	38.1	19.3	95.1
	$\tilde{\psi}$	128	1.1	43.5	53.2	55.3	95.9
		256	1.7	29.6	51.0	37.4	95.7
		512	1.0	20.5	50.4	25.9	95.4
		1024	0.6	14.4	50.2	18.1	95.2
20	$\hat{\psi}$	128	136.5	139.5	5.5	166.6	94.2
		256	61.6	67.6	13.2	83.6	92.0
		512	29.3	37.4	21.8	47.1	92.8
		1024	14.3	22.8	29.1	28.9	93.6
	$\tilde{\psi}$	128	-18.0	63.1	63.0	78.3	95.4
		256	-1.8	40.9	54.1	51.4	96.2
		512	0.4	28.0	51.4	35.2	95.8
		1024	0.5	19.5	50.5	24.5	95.4
30	$\hat{\psi}$	128	260.6	261.1	1.1	294.8	91.7
		256	110.9	113.7	5.6	133.9	86.2
		512	51.7	57.2	13.3	70.0	88.5
		1024	24.9	32.2	21.8	40.2	91.2
	$\tilde{\psi}$	128	-70.7	98.1	79.3	116.7	91.2
		256	-10.5	52.4	59.0	65.3	96.2
		512	-1.2	35.1	52.9	44.1	96.1
		1024	0.2	24.4	51.0	30.6	95.7

Closed-form expressions for the quantities required in expression (9) are cumbersome for this model because the higher-order likelihood quantities involve derivatives of $C(\boldsymbol{\theta})$. In contrast, simulation from the model is straightforward using a bivariate rejection sampler with a uniform proposal density (Mardia *et al.*, 2007), as implemented in `rvmsin()` from BAMB1. We therefore estimate $\mathbf{V}(\boldsymbol{\theta})$, $\mathbf{P}_t(\boldsymbol{\theta})$ and $\mathbf{Q}_t(\boldsymbol{\theta})$ by Monte Carlo simulation from the fitted model, as described in Section 5. Specifically, at $\hat{\boldsymbol{\theta}}$ we simulate single observations from the fitted bivariate von Mises sine model and use the numerical gradient and Hessian of the corresponding single-observation log-likelihood contribution to estimate the required quantities. The gradient $\dot{\psi}$ and Hessian $\ddot{\psi}$ of $h(\boldsymbol{\theta})$ are also computed numerically. Substituting these quantities into (9) gives the median bias-corrected estimator $\tilde{\psi}$ of the circular variance. The mean bias-corrected estimator is computed as $\hat{\psi} - \hat{b}_{(\psi)}$.

Table S9: Comparison of the plug-in estimator $\hat{\psi} = h(\hat{\boldsymbol{\theta}})$ of the circular variance of the first angular component in the bivariate von Mises sine model with the mean bias-corrected focus estimator $\hat{\psi} - \hat{b}_{(\psi)}$ and the MBC focus estimator $\tilde{\psi}$, both based on Monte Carlo estimates of $\mathbf{V}(\boldsymbol{\theta})$, $\mathbf{P}_1, \dots, \mathbf{P}_p$ and $\mathbf{Q}_1, \dots, \mathbf{Q}_p$ at $\hat{\boldsymbol{\theta}}$ ($R = 500$), in terms of simulation-based estimates of probability of underestimation (PU), mean bias (BIAS), mean absolute deviation (MAD), root mean squared error (RMSE), and coverage of nominally 95% Wald-type confidence intervals based on supplied and compatible estimates of $\boldsymbol{\theta}$ (Wald [S] and Wald [C], respectively), and HulC-type confidence intervals. All summaries are $\times 100$.

n	Estimator	BIAS	MAD	PU	RMSE	Wald	HulC
30	$\hat{\psi}$	-0.8	4.9	57.9	6.1	90.3	88.9
	$\hat{\psi} - \hat{b}_{(\psi)}$	0.1	5.1	53.2	15.5	91.0	92.9
	$\tilde{\psi}$	0.5	5.2	50.2	15.6	91.5	94.0
60	$\hat{\psi}$	-0.3	3.5	54.3	4.4	91.9	91.7
	$\hat{\psi} - \hat{b}_{(\psi)}$	0.0	3.5	51.5	4.4	92.5	94.2
	$\tilde{\psi}$	0.2	3.5	49.3	4.4	92.5	95.6
90	$\hat{\psi}$	-0.2	2.8	55.2	3.5	93.4	92.9
	$\hat{\psi} - \hat{b}_{(\psi)}$	0.0	2.8	52.1	3.5	93.9	94.7
	$\tilde{\psi}$	0.2	2.8	50.2	3.5	94.0	94.9
120	$\hat{\psi}$	-0.2	2.4	54.2	3.0	94.3	93.4
	$\hat{\psi} - \hat{b}_{(\psi)}$	0.0	2.4	51.7	3.0	94.4	94.7
	$\tilde{\psi}$	0.1	2.4	49.9	3.0	94.6	95.0

Table S9 compares the estimation performance of $\hat{\psi}$, $\hat{\psi} - \hat{b}_{(\psi)}$, and $\tilde{\psi}$, and the coverage properties of confidence intervals based on them, using 5000 simulated samples for each $n \in \{30, 60, 90, 120\}$ from the bivariate von Mises sine model with $\mu_1 = \pi/3$, $\mu_2 = 2\pi/3$, $e^{\tau_1} = 2.5$, $e^{\tau_2} = 3$, and $\lambda = 1$. For each simulated sample, the quantities required by the focused corrections are estimated using $R = 500$ Monte Carlo samples from the fitted model. We report Wald-type confidence intervals based on supplied estimates of $\boldsymbol{\theta}$, and HulC-type confidence intervals. The median bias-corrected estimator has probability of underestimation closer to 50% than the maximum likelihood estimator, at the cost of slightly larger root mean squared error. The Wald-type intervals perform similarly and tend to undercover, particularly at smaller sample sizes. By contrast, HulC-type intervals based on $\tilde{\psi}$ attain coverage closer to the nominal level, substantially outperforming the corresponding intervals based on $\hat{\psi}$.

References

- Chakraborty S, Wong SWK (2021). “BAMBI: An R package for fitting bivariate angular mixture models.” *Journal of Statistical Software*, **99**, 1–69. doi:10.18637/jss.v099.i11.
- Firth D (1993). “Bias reduction of maximum likelihood estimates.” *Biometrika*, **80**(1), 27–38. doi:10.1093/biomet/80.1.27.
- Mardia KV, Frelsen J (2012). “Statistics of bivariate von Mises distributions.” In T Hamelryck, KV Mardia, J Ferkinghoff-Borg (eds.), “Bayesian Methods in Structural Bioinformatics,” chapter 6. Springer, Berlin, Heidelberg. doi:10.1007/978-3-642-27225-7_6.
- Mardia KV, Taylor CC, Subramaniam GK (2007). “Protein bioinformatics and mixtures of bivariate von Mises distributions for angular data.” *Biometrics*, **63**, 505–512. doi:10.1111/j.1541-0420.2006.00682.x.
- McCullagh P (2018). *Tensor Methods in Statistics*. Chapman and Hall/CRC, Boca Raton. doi:10.1201/9781351077118.
- Pace L, Salvan A (1997). *Principles of Statistical Inference from a Neo-Fisherian Perspective*. World Scientific, Singapore. doi:10.1142/3409.
- Singh H, Hnizdo V, Demchuk E (2002). “Probabilistic model for two dependent circular variables.” *Biometrika*, **89**(3), 719–723. doi:10.1093/biomet/89.3.719.